Dataset
Amazon Reviews 2023 (Video_Games tier)
262,547 raw rows
replicated on Musical_Instruments

Data Ingestion, Cleaning & Leakage-Safe Split -- Modules 1-3, 22

MODULE 1
Ingestion & Schema Adapter
field presence checked, never invented

MODULE 2
Cleaning & k-core Filter
262,547 -> 144,178 rows (54.9% retained)

MODULE 22
Chronological Split -- 80/10/10
train 115,342 * val 14,418 * test 14,418

ENFORCED
Leakage Guards
6 assertions in tests/ -- no val/test
edge ever enters the training graph
terminal -- a check on the split,
not a data-producing stage

MODULES 3-4
EDA * Outlier Analysis
18.7% corroborated outliers (>=2 detectors agree) --
flagged, never deleted, so the trust detectors have evidence to find
terminal -- informs design decisions and the written
report (Sec. G); nothing downstream consumes it

(a) Feature Engineering -- Sentiment, Semantics & Fusion -- Modules 5-8

review_title + review_text

MODULE 6
BERT Sentiment Encoder
nlptown/bert-base-multilingual-
uncased-sentiment -> pos/neg/neu/compound

MODULE 5
5 Structured Features
length * helpfulness * verified *
recency * rating-centred

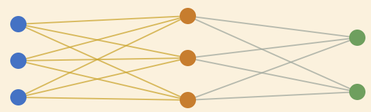
MODULE 7
SBERT Semantic Encoder
all-MiniLM-L6-v2 * mean-pool * L2-norm
-> 384-d, unit-norm

MODULE 8
Semantic-Behavioural Fusion Layer
concat -> LayerNorm ->
Linear(390->256) -> GELU ->
Dropout -> Linear(256->128)
ablated: no gain over concatenation;
the best variant bypasses it

concat arm (mode="concat") kept alongside the learned projection so "does fusion beat concatenation?" is answered, not assumed. Measured: it does not (Table V).

(b) Trust-Aware Heterogeneous Graph -- Modules 9, 11-13

5-type heterogeneous graph G = (V, E)



138,480 nodes * 718,506 edges (incl. reverses)
user (blue) / product (orange) / brand-category (green)

MODULE 11
Behavioural Trust Detector
indicators: shared text * repeats * bursts *
paraphrase match (sentence embeddings)

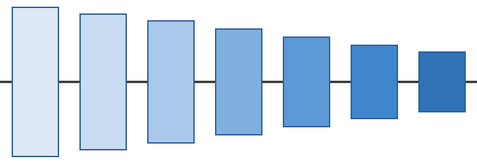
MODULE 12
Anomaly -> Trust
tau = 1 - rank-norm(score), clamp [0.05, 1];
flagged accounts are gated out of the graph

MODULE 13
Temporal Recency Weight
exp(-lambda*age), lambda=2 (tuned on validation), floor 0.01
age from max(train.timestamp), never now

OUTPUT
Trust x Recency Sample Weights
w_e = base * tau_e^a * recency_e^b scales each BPR term
(HGTConv cannot read an edge weight)

Heterogeneous Graph Transformer (HGT) Encoder -- Module 14

Type-Specific Linear Projection -> 128-d
per node type: user 20-d, product 18-d,
review 390-d, brand/category learned



HGT: 2 layers x 4-head relational attention, full-batch

OUTPUT
User / Product Embeddings
128-d, score(u,i) = e_u . e_i

MODULE 20/31
BPR Objective, AdamW, Edge-Masked Training
each epoch 50% of interaction edges are hidden from message passing
and become the targets; validation NDCG@10 early-stops and restores the best epoch

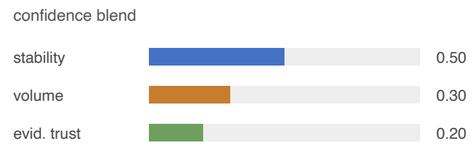
MODULE 15
Recommendation Scoring
Full-catalogue ranking, positives masked;
last r slots: new products by text

MODULE 16
Explanation Branch
6 evidence sources; each cited fact is
tested by removing it and re-running

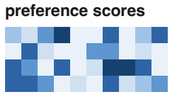
MODULE 17
Counterfactual Branch
Bisection search over one perturbable
factor -> magnitude to enter Top-K, or infeasible

MODULE 18
Confidence Branch
evidence sufficiency 0.5 / 0.3 / 0.2, plus
a separately calibrated hit probability

confidence blend



preference scores



y_u,i score map (illustrative)

Trust-Calibrated Explainable Top-K Recommendation

Live example (real run, un-curated: the evaluable user with the longest history): 'Sega Dreamcast Console' recommended to a user with 117 training interactions and trust 0.996. 9 high-trust users with overlapping history also bought it, it belongs to legacy systems, which dominates the user's history, and the user's recent activity is concentrated in legacy systems. Held-out outcome: the top item was not a hit in the test split.

MODULE 39
Streamlit Dashboard
reads generated artefacts only --
trains nothing, computes no metric of its own